

Database Systems

CSE 215

Lecture 04

2022

Queries Over Multiple Table

Problems with single table

Add company info like country, stock price

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Problems with single table

Product

PName	Price	Category	Manufacturer	Country	StockPrice
Gizmo	\$19.99	Gadgets	GizmoWorks	USA	25
Powergizmo	\$29.99	Gadgets	GizmoWorks	USA	25
SingleTouch	\$149.99	Photography	Canon	Japan	65
MultiTouch	\$203.99	Household	Hitachi	Japan	15

Solution to the problem: Split

Product

PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Company

Manufacturer	Country	StockPrice
GizmoWorks	USA	25
GizmoWorks	USA	25
Canon	Japan	65
Hitachi	Japan	15

Solution to the problem: Split

Product

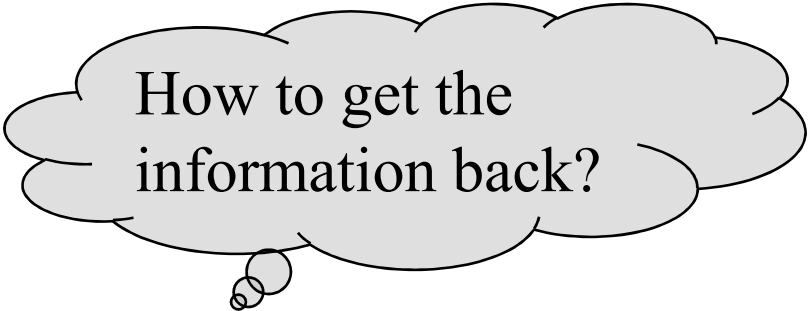
PName	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Foreign
key

Company

Manufacturer	Country	StockPrice
GizmoWorks	USA	25
Canon	Japan	65
Hitachi	Japan	15

Primary
Key



How to get the
information back?

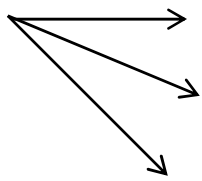
Cross Product of two tables

Product $A \times B$

Part of the SELECT statement – list more than one table after keyword FROM.

```
SELECT A, R.B, S.B, C
FROM R, S;
```

<i>R</i>		<i>S</i>	
A	B	B	C
-----		-----	
1	x	x	5
2	y	y	6
		z	7



$R \times S$

A	R.B	S.B	C

1	x	x	5
1	x	y	6
1	x	z	7
2	y	x	5
2	y	y	6
2	y	z	7

Join

```
SELECT *
FROM R, S
WHERE R.B=S.B;
```

<i>R</i>		<i>S</i>	
A	B	B	C
-----		-----	
1	x	x	5
2	y	y	6
		z	7

$R \times S$

A	R.B	S.B	C

1	x	x	5
1	x	y	6
1	x	z	7
2	y	x	5
2	y	y	6
2	y	z	7

Join

```
SELECT *  
FROM R, S  
WHERE R.B=S.B;
```

<i>R</i>	
A	B

1	x
2	y

<i>S</i>	
B	C

x	5
y	6
z	7

A	R.B	S.B	C

1	x	x	5
2	y	y	6

Join

R natural join S

```
SELECT A, R.B, C
FROM R, S
WHERE R.B=S.B;
```

SELECT *
FROM R NATURAL JOIN S

R natural join S

<i>R</i>	
A	B

1	x
2	y

<i>S</i>	
B	C

x	5
y	6
z	7

A	R.B	C

1	x	5
2	y	6

Meaning (Semantics) of SQL Queries Using Cross Product

```
SELECT a1, a2, ..., ak  
FROM R1, R2, ..., Rn  
WHERE Conditions
```

```
Answer = {}  
for x1 in R1 do  
  for x2 in R2 do  
    .....  
    for xn in Rn do  
      if Conditions  
        then Answer = Answer + {(a1, ..., ak)}  
return Answer
```

Meaning (Semantics) of SQL Queries Using Cross Product

```
SELECT DISTINCT  $a_1, a_2, \dots, a_k$   
FROM  $R_1, R_2, \dots, R_n$   
WHERE Conditions
```

```
Answer = {}  
for  $x_1$  in  $R_1$  do  
  for  $x_2$  in  $R_2$  do  
    .....  
    for  $x_n$  in  $R_n$  do  
      if Conditions  
        then Answer = Answer  $\cup \{(a_1, \dots, a_k)\}$   
return Answer
```

Joins

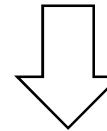
Product

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Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Company

Cname	StockPrice	Country
GizmoWorks	25	USA
Canon	65	Japan
Hitachi	15	Japan

```
SELECT *
FROM Product, Company
WHERE manufacturer=cname
```



Product join Company

PName	Price	Category	Manufacturer	Cname	StockPrice	Country
Gizmo	\$19.99	Gadgets	GizmoWorks	GizmoWorks	25	USA
Powergizmo	\$29.99	Gadgets	GizmoWorks	GizmoWorks	25	USA
SingleTouch	\$149.99	Photography	Canon	Canon	65	Japan
MultiTouch	\$203.99	Household	Hitachi	Hitachi	15	Japan

Joins

Product (pname, price, category, manufacturer)

Company (cname, stockPrice, country)

Find all products under \$200 manufactured in Japan;
return their names and prices.

```
SELECT pname, price
FROM Product, Company
WHERE manufacturer=cname
AND country='Japan'
AND price < 200
```



Join
between Product
and Company

Joins

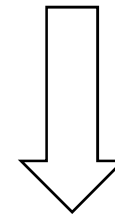
Product

PName	Price	Category	Manufacturer
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Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Company

Cname	StockPrice	Country
GizmoWorks	25	USA
Canon	65	Japan
Hitachi	15	Japan

```
SELECT pname, price
FROM Product, Company
WHERE manufacturer=cname
AND country='Japan'
AND price < 200
```



PName	Price
SingleTouch	\$149.99

More Joins

Product (pname, price, category, manufacturer)

Company (cname, stockPrice, country)

Find all Chinese companies that manufacture electronic products or toy.

```
SELECT cname
FROM Product, Company
WHERE manufacturer = cname
AND country = 'China'
AND (category = 'electronic' OR category = 'toy')
```


A Subtlety about Joins

Product (pname, price, category, manufacturer)

Company (cname, stockPrice, country)

Find all countries that manufacture some product in the
'Gadgets' category.

A Subtlety about Joins

Product

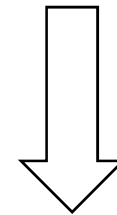
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Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

Company

<u>Cname</u>	StockPrice	Country
GizmoWorks	25	USA
Canon	65	Japan
Hitachi	15	Japan

```
SELECT country
FROM Product, Company
WHERE manufacturer=cname
AND category='Gadgets'
```

What is
the problem ?



Country
??
??

A Subtlety about Joins

Product

<u>Name</u>	Price	Category	Manufacturer
Gizmo	\$19.99	Gadgets	GizmoWorks
Powergizmo	\$29.99	Gadgets	GizmoWorks
SingleTouch	\$149.99	Photography	Canon
MultiTouch	\$203.99	Household	Hitachi

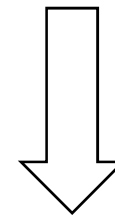
Company

<u>Cname</u>	StockPrice	Country
GizmoWorks	25	USA
Canon	65	Japan
Hitachi	15	Japan

```
SELECT country
FROM Product, Company
WHERE manufacturer=cname
AND category='Gadgets'
```

Duplicates !
What's the
solution ?

Use 'distinct' keyword



Country
USA
USA

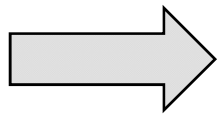
Tuple Variables

Person(pname, address, worksfor)

Company(cname, address)

```
SELECT  pname, address
FROM    Person, Company
WHERE   worksfor = cname
```

Which
address ?



```
SELECT  Person.pname, Company.address
FROM    Person, Company
WHERE   Person.worksfor = Company.cname
```



```
SELECT  x.pname, y.address
FROM    Person AS x, Company AS y
WHERE   x.worksfor = y.cname
```

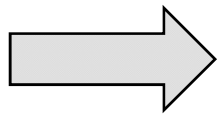
Tuple Variables

Person(pname, address, worksfor)

Company(cname, address)

```
SELECT pname, address
FROM   Person, Company
WHERE  worksfor = cname
```

Which
address ?



```
SELECT Person.pname, Company.address
FROM   Person, Company
WHERE  Person.worksfor = Company.cname
```



```
SELECT x.pname, y.address
FROM   Person AS x, Company AS y
WHERE  x.worksfor = y.cname
```

No 'AS' in
Oracle

Examples Examples ...

Product (maker, model, type)

PC (model, speed, ram, hd, price)

Laptop (model, speed, ram, hd, screen, price)

Find the model number and price of all PCs and Laptops made by manufacturer 'B'

Find those manufacturers that sell both PCs and Laptops

Find those manufacturers that sell Laptops, but not PCs

Union

Union $A \cup B$

Use SQL keyword UNION. Tables must be compatible ... have the same attributes (column headings).

Product (maker, model, type)

PC (model, speed, ram, hd, price)

Laptop (model, speed, ram, hd, screen, price)

Find the model number and price of all PCs and Laptops made by manufacturer 'B'

```
SELECT PC.model, PC.price
FROM Product, PC
WHERE Product.model = PC.model
AND Product.maker = 'B'

UNION

SELECT Laptop.model, Laptop.price
FROM Product, Laptop
WHERE Product.model = Laptop.model
AND Product.maker = 'B'
```


Intersection

Intersection $A \cap B$

Use SQL keyword `INTERSECT`. Tables must be compatible.

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find those manufacturers that sell both PCs and Laptops

```
SELECT maker  
FROM Product  
WHERE type = 'pc'
```

INTERSECT

```
SELECT maker  
FROM Product  
WHERE type = 'laptop'
```

Difference

Use SQL keyword **MINUS**. Tables must be compatible.

Product (maker, model, type)
PC (model, speed, ram, hd, price)
Laptop (model, speed, ram, hd, screen, price)

Find those manufacturers that sell Laptops, but not PCs

```
SELECT maker  
FROM Product  
WHERE type = 'laptop'
```

MINUS

```
SELECT maker  
FROM Product  
WHERE type = 'pc'
```

Unlike SELECT operator,

UNION, INTERSECT, MINUS removes duplicates.

Thus they convert bags to sets first and then return *sets* and not *Bags*

Example:

T1: 1, 2, 2, 2, 3, 4, 4

T2: 2, 3, 4, 4, 4, 5

SELECT * FROM T1

UNION (or INTERSECT or MINUS operator)

SELECT * FROM T2

T1 converted to set: 1, 2, 3, 4

T2 converted to set : 2, 3, 4, 5

T1 UNION T2: 1, 2, 3, 4, 5

T1 INTERSECT T2: 2, 3, 4

T1 MINUS T2: 1

If you want duplicates you must use the keyword **ALL**.

Example:

T1: 1, 2, 2, 2, 3, 4, 4

T2: 2, 3, 4, 4, 4, 5

SELECT * FROM T1

UNION ALL (or INTERSECT or MINUS)

SELECT * FROM T2

T1 UNION ALL T2: 1, 2, 2, 2, 2, 3, 3, 4, 4, 4, 4, 4, 5

T1 INTERSECT ALL T2: 2, 3, 4, 4

T1 EXCEPT ALL T2: 1, 2, 2

For example

```
SELECT maker  
FROM Product  
WHERE type = 'pc'  
  
UNION ALL  
  
SELECT maker  
FROM Product  
WHERE type = 'laptop'
```

If a maker appears a times in the first set and b times in the second set then it appears $(a+b)$ times in the final set when UNION ALL is used.

For example

```
SELECT maker  
FROM Product  
WHERE type = 'pc'
```

INTERSECT ALL

```
SELECT maker  
FROM Product  
WHERE type = 'laptop'
```

If a maker appears a times in the first set and b times in the second set then it appears $\text{MIN}(a,b)$ times in the final set when INTERSECT ALL is used.

For example

```
SELECT maker  
FROM Product  
WHERE type = 'pc'  
  
EXCEPT ALL  
  
SELECT maker  
FROM Product  
WHERE type = 'laptop'
```

If a maker appears a times in the first set and b times in the second set then it appears $\text{MAX}(0, a-b)$ times in the final set when EXCEPT ALL is used.

For example

ORACLE doesn't support INTERSECT ALL and MINUS ALL
ONLY UNION ALL is supported.